SOP: Centrifugal Slurry Pump P-201A/B Operation & Maintenance

1. Mechanical Architecture and Technical Specifications

Slurry Pump P-201A (Primary) and P-201B (Standby) are heavy-duty centrifugal single-stage process pumps operating in the Fluid Catalytic Cracking Unit (FCCU). The pumps transfer catalyst slurry oil from the main fractionator bottom sumps via suction line 8"-P-203-CS-150 and discharge line 6"-P-204-CS-150.
Technical Baseline Data:
- Rated Flow: 250.0 m3/hr | Total Dynamic Head (TDH): 45.0 m | Slurry Density: 1050.0 kg/m3
- Motor Rating: 55.0 kW | Synchronous Speed: 2950 RPM | Voltage: 415 V, 3-Phase, 50 Hz
- Shaft Seal Configuration: Dual mechanical cartridge seal with API Plan 53B pressurized barrier system.

2. API Plan 53B Seal Flush and Pre-Start Alignment

Prior to starting pump P-201A or standby P-201B, verify the mechanical seal barrier fluid system:
a) Inspect the Plan 53B bladder accumulator pressure gauge: barrier fluid pressure must read exactly 3.5 bar gauge (minimum 1.5 bar above maximum expected stuffing box pressure of 2.0 bar gauge).
b) Verify lube oil level in the bearing housing sight glass is maintained between 45% and 55% level.
c) Recommended lubricant: ISO VG 46 synthetic turbine oil with high thermal stability.
d) Fully open suction isolation valve on line 8"-P-203-CS-150. Ensure pump casing is vented and primed.
e) Crack open discharge throttling valve on line 6"-P-204-CS-150 to approximately 10% open position.

3. Operating Temperature and ISO 10816-3 Vibration Limits

Continuous online condition monitoring instrumentation provides real-time telemetry to the DCS:
- Bearing Housing Temperature Limits:
  * Normal Steady-State: 50 deg C to 65 deg C
  * High Temperature Warning Advisory Alarm: 75.0 deg C
  * Emergency High-High Automated Trip Threshold: 85.0 deg C

- ISO 10816-3 Category Class II Vibration Thresholds (Broadband Velocity RMS):
  * Zone A/B (Normal Unrestricted Continuous Operation): Vibration < 2.8 mm/s RMS
  * Zone C (High Vibration Warning Alarm Threshold): 7.1 mm/s RMS (Trigger inspection within 2 hours)
  * Zone D (Emergency High-High Trip Threshold): 9.0 mm/s RMS (Immediate automated trip shutdown)

SOP: Centrifugal Slurry Pump P-201A/B - Operational Sequencing

4. Primary-to-Standby Pump Auto-Switchover Sequence

When high vibration (>7.1 mm/s RMS) or bearing temperature (>75 deg C) occurs on primary pump P-201A, execute the bumpless switchover procedure to standby pump P-201B as follows:
Step 1: Check P-201B auxiliary systems: verify lube oil level and API Plan 53B barrier pressure >= 3.5 bar.
Step 2: Confirm P-201B suction valve is 100% open and minimum flow recirculation bypass valve is open.
Step 3: Start P-201B electric motor from DCS console or field emergency start button.
Step 4: Observe discharge pressure transmitter PT-204: confirm pressure reaches 5.2 bar gauge within 6 sec.
Step 5: Gradually open P-201B discharge valve to 100% over 30 seconds while simultaneously throttling P-201A discharge valve to 0% to avoid flow surging or hydraulic hammer in discharge line 6"-P-204-CS-150.
Step 6: Stop P-201A motor. Apply lockout/tagout (LOTO) padlocks on electrical breaker P-201A-MCC.

5. Darcy-Weisbach Friction and NPSH Verification Formulas

Hydraulic friction loss across discharge line 6"-P-204-CS-150 is governed by the Darcy-Weisbach equation:
Head Loss: h_f = f * (L / D) * (V^2 / (2 * g))
Where pipe diameter D = 0.15 m, total equivalent length L = 85.0 m, fluid velocity V = 2.2 m/s, Darcy friction factor f = 0.018, and gravitational acceleration g = 9.81 m/s^2.
Calculated head loss h_f = 2.51 m, corresponding to pressure drop Delta P = 25.83 kPa.

Net Positive Suction Head (NPSH) Security Margin:
NPSH_Available = 4.2 m, NPSH_Required = 2.8 m. Safety margin = 1.4 m (Mandatory margin >= 1.0 m). If suction strainer differential pressure reaches 0.35 bar, clean strainer immediately to prevent cavitation.